9 **(a)** Cosmic microwave background radiation (CMBR) can be observed in every direction in space around us.

(i) State when CMBR was produced.

..... [1]

(ii) State and explain how the wavelength of this radiation has changed since it was formed.

.....

.....

..... [2]

(b) Table 9.1 shows data about planets in our solar system.

Table 9.1

planet	average radius of orbit around the Sun / 10^6 km	time for one orbit / Earth years	average surface temperature / °C
Venus	108	0.62	460
Earth	150	1.00	15
Mars	228	1.90	–65
Jupiter	778	12.00	–110
Saturn	1430	29.00	–140

(i) The orbit of Venus is approximately circular.

Calculate the average orbital speed v of Venus. Give your answer in km / h.

orbital speed = ... km / h [3]

(ii) State how the average surface temperature of a planet changes as its distance from the Sun increases.

..... [1]

[Total: 7]

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